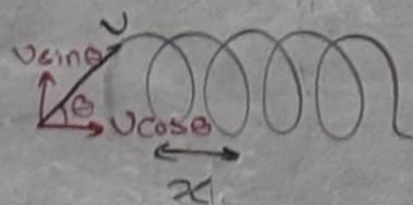


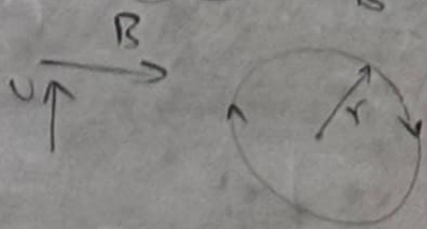
سجہ الٹرا وائے

straight: $F_B = 0 \rightarrow$

Helix: 

$$\lambda = v \cos \theta = \frac{2\pi m}{qB}$$

circular: $F_B = \text{max} \rightarrow F_B = qvB$



$$qBr = mv$$

$$r = \frac{mv}{qB}$$

$$* v = \sqrt{\frac{2KE}{m}} \quad \text{or} \quad v = \sqrt{\frac{2qV_p}{m}}$$

$$* T = \frac{2\pi m}{qB}$$

$$* v = \frac{F_B}{qB} \quad \text{or} \quad B = \frac{F_B}{qv}$$

Electromagnetic wave

electromagnetic wave consists of E and B .

E and B are Perpendicular

E and H are Perpendicular

characteristics

① electromagnetic wave travels:

① in free space: it has speed of light $\rightarrow c = 3 \times 10^8 \text{ m/s}$

② in medium: it has less than $c = 3 \times 10^8$ $v < c$

more density
 $\downarrow$
light moves slower
less density
 $\downarrow$
light moves faster

$v = ?\% \times 3 \times 10^8$
60% of speed of light: $v = \frac{60}{100} \times 3 \times 10^8 = 1.8 \times 10^8 \text{ m/s}$

80% of speed of light: $v = \frac{80}{100} \times 3 \times 10^8 = 2.4 \times 10^8 \text{ m/s}$

③ energy of EM depends directly on frequency
more energy $\rightarrow$ more frequency

energy of EM depends inversely on wavelength
more energy $\rightarrow$ short wavelength

④ chargeless $\rightarrow$?

⑤ Transverse $\rightarrow$?

بسم الله الرحمن الرحيم

Electromagnetic Wave

electromagnetic Spectrum:

الطيف الكهرومغناطيسي
المرئي

- ① Radio waves: longest wavelength $\rightarrow$ lowest energy
- ② Microwaves: short waves used in Radar systems
- ③ InfraRed rays: Produced from hot body
- ④ Visible light: Light detected by human eyes $\rightarrow$

1	2	3	4	5	6	7
---	---	---	---	---	---	---

Red (Colors) Violet
- ⑤ UltraViolet rays: From sun and causes harmful $\rightarrow$ Ozone converts UV into IR
- ⑥ X-rays: Produced from deceleration of high speed electron causes damage
- ⑦ Gamma rays: shortest wavelength $\rightarrow$ highest energy causes damage

Difference between wave and Ray

wavelengths:

m	10^3 m
mm	10^6 m
μ m	10^9 m
nm	10^{12} m

frequencies:

Hz	10^3 Hz
KHz	10^6 Hz
MHz	10^9 Hz
GHz	10^{12} Hz